



Programming Fundamentals — Assignment

Lab 03: Decision Making in C++ (Selection Statements)

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Github Repo: <https://github.com/aaliyann-exe/programming-fundamentals-lab>

Task 1: Write a program that takes two integers from the user and displays the result of all six relational operators (use boolalpha so the output shows true / false).

Output:

```
Enter first number: 6
Enter second number: 7
6 == 7 : false
6 != 7 : true
6 > 7 : false
6 < 7 : true
6 >= 7 : false
```

Task 2: Take the bill amount from the user. If the bill is more than Rs. 5000 give a 10% discount. Print the discount (only when it applies) and the amount payable.

Output:

```
Enter bill amount: 6700
Discount: 670
Amount payable: 6030
```

Task 3: Take the age of a person. If the age is 18 or more, print that they are eligible to apply for a driving licence; otherwise print how many more years they must wait.

Output:

Above 18:

```
Enter your age: 21
You are eligible.
```

Below 18:

```
Enter your age: 13
You are not eligible yet. Please wait 5 more year(s).
```

Task 4: The correct user ID is the string cs101 and the correct password is the number 1234. Take both from the user. Using a nested if, first check the ID; only if it is correct check the password. Print "Wrong user ID", "Wrong password" or "Login successful" accordingly.

Output:

Successful Login:

```
Enter user ID: cs101
Enter password: 1234
Login successful
```

Unsuccessful Login:

```
Enter user ID: idk12
Enter password: umm67
Wrong user ID and password
```

Task 5: Take the units consumed and calculate the bill. The rate per unit depends on the bracket the total units fall in and applies to all units: up to 100 units = Rs. 10, 101 to 300 units = Rs. 15, above 300 units = Rs. 22. If the units are negative, print "Invalid units".

Pseudocode:

START

INPUT units

IF units < 0 THEN

PRINT "Invalid units"

ELSE

IF units <= 100 THEN

rate = 10

ELSE IF units <= 300 THEN

rate = 15

ELSE

rate = 22

END IF

bill = units * rate

PRINT "Rate per unit: Rs. ", rate

PRINT "Total bill: Rs. ", bill

END IF

END

Output:

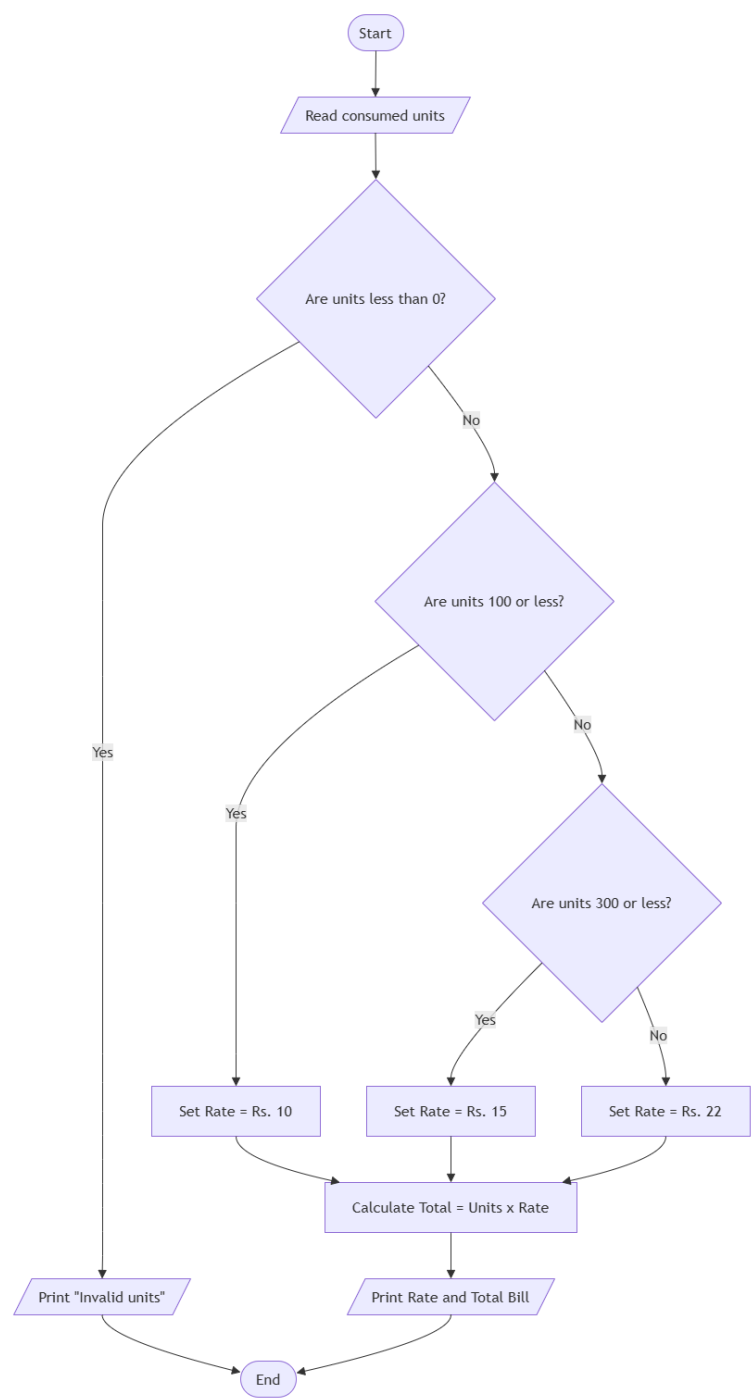
Correct Units:

```
Enter units consumed: 320
Rate per unit: Rs. 22
Total bill: 7040
```

Negative Units:

```
Enter units consumed: -20
Invalid units
```

Flowchart:



Task 6: Take a year from the user and tell whether it is a leap year. A year is a leap year if it is divisible by 4 and not divisible by 100, or it is divisible by 400. Test with 2024, 1900 and 2000.

Output:

Leap Year:

```
Enter year: 2024
2024 is a leap year.
```

Normal Year:

```
Enter year: 2026
2026 is not a leap year.
```

Task 7: Take two numbers and an operator (+, -, * or /) from the user. Use a switch on the operator to print the result. For / with a second number of 0 print "Cannot divide by zero", and for any other operator print "Invalid operator".

Output:

Correct Operator and Non-Zero:

```
Enter first number: 6
Enter operator (+, -, *, /): -
Enter second number: 7
Result: -1
```

Dividing by Zero:

```
Enter first number: 20
Enter operator (+, -, *, /): /
Enter second number: 0
Cannot divide by zero
```

Invalid Operator:

```
Enter first number: 49
Enter operator (+, -, *, /): &
Enter second number: 2
Invalid operator
```

Task 8: A cinema counter needs a program that works out the amount to pay. Students are given the scenario only, not the code, and must (1) write the pseudocode, (2) draw the flowchart, then (3) implement it. The program uses a switch, logical operators and if-else-if together.

Pseudocode:

START

INPUT units

IF units < 0 THEN

PRINT "Invalid units"

ELSE

IF units <= 100 THEN

rate = 10

ELSE IF units <= 300 THEN

rate = 15

ELSE

rate = 22

END IF

bill = units * rate

PRINT "Rate per unit: Rs. ", rate

PRINT "Total bill: Rs. ", bill

END IF

END

Output:

Test Run:

```
Enter ticket class (1 = Economy, 2 = Gold, 3 = Platinum): 2
Enter your age: 20
Are you a student? (y/n): y
Enter number of tickets: 3
Total amount to pay: Rs. 1920
```

Flowchart:

